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A comparison of visual- and stereo-video based fish community assessment methods in tropical and temperate marine waters of Western Australia

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Abstract

Robust assessments of abundance and diversity are essential components of research programs aimed at detecting changes in marine fish assemblages through space and time. This study examined the comparability of Underwater Visual Census (UVC), and Diver Operated stereo-Video (Stereo-DOV) datasets collected across a wide latitudinal range (15°) on coral reefs and temperate rocky shores. There were some differences between methods with regards to both species richness and abundance, with UVC consistently recording higher measures of species richness. Differences were most pronounced at tropical locations where these measures were high. Differences in the characteristics of fish assemblages were primarily driven by UVC differentiating between scarid and pomacentrid species and detecting more cryptic species. When examined at higher taxonomic or functional levels however, there was greater comparability between the assemblages recorded by each method, particularly in temperate locations. The UVC method also recorded higher abundances of species targeted by fishers in tropical/sub-tropical locations, and subsequently obtained a much higher proportion of length measurements for these species. Data collected using stereo-DOV took 2-3 times longer to obtain than with UVC due to extensive post-processing time required by the stereo-DOV method. This study shows that data collected by the two methods are most comparable in temperate locations, or when examined at higher taxonomic/functional levels. Comparisons should however be more cautious in higher diversity locations, or when assessing at finer taxonomic resolution. When assessing the suitability of either method, availability of time, funding, and relevant expertise should be primary considerations.

A central issue in the development of marine research and monitoring programs is the identification of robust methodologies aimed at detecting spatial and temporal changes in abundance, diversity, and biomass of faunal assemblages (e.g., Maxwell and Jennings 2005; Cheal et al. 2008; Murphy and Jenkins 2010). Decisions on how to assess these metrics are initially based on the type of information required, what specific indices must be measured, the level of precision required to detect change, the repeatability of the method, and the environmental conditions under which assessments will take place (Connell et al. 1998; Willis et al. 2000; Smale et al. 2011). Importantly, the human capital and financial resources available to collect data must also be considered, as costs associated with different data collection techniques can vary substantially (Langlois et al. 2010). However, for long-term monitoring projects technological advancements, scientific innovations, sporadic funding, and changes in the availability of staff expertise often mean that the most suitable assessment methods change through time (Murphy and Jenkins 2010). This may result in time-series datasets that are collected using several techniques. To ensure that trends in data are not attribut-

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able to changes in methodology it is imperative that the compatibility of data collected using these methods is assessed (e.g., Cheal and Thompson 1997). Ideally, this should occur before changes in methods are implemented.

In shallow marine ecosystems assessing the condition of fish assemblages is frequently considered a priority given the high economic, social, ecological, and conservation value placed on many species. A wide variety of methods and techniques can be used to assess the condition of fish assemblages. Each of these methods has specific advantages and disadvantages with respect to the type of fish data they are capable of collecting, along with the relative cost associated with collecting that data (time, financial, and expertise).

The most common method of surveying and monitoring shallow water fish assemblages is underwater visual census (UVC) (Caldwell 2011). This technique is popular as it requires minimal equipment that is cheap and easily accessible (commonly just a pencil, slate, and tape measure), and does not involve significant time investment post data collection in the field. However, it does generally require a high degree of observer expertise, due to the knowledge required to rapidly identify fishes in situ. Despite this, UVC has been commonly employed by fish monitoring programs around the world, particularly within research institutions where appropriate expertise is readily available (e.g., Halford and Thompson 1994). Consequently, much of the nonextractive historical data on shallow water fish abundance and richness worldwide has been collected using the UVC method.

Underwater video camera techniques have increasingly been used for assessing shallow water fish assemblages in more recent times (e.g., Bortone et al. 1991; Parker et al. 1994; Moser et al. 1998), and the continued development of camera technology has enabled rapid advances in these techniques (e.g., image resolution, storage capacity, software capabilities; Shortis et al. 2009). Using two cameras set in a stereo configuration enables precise estimates of fish lengths, which allows robust evaluation of size distributions and biomass where length/weight relationships are known (Harvey et al. 2002). When pushed through the water by a diver, this stereo–Diver Operated Video (stereo-DOV) method can be used to record fish assemblages within defined and accurate sample unit boundaries (Harvey et al 2004). This is generally thought to be advantageous over UVC diver observations in that video footage can be repeatedly analyzed on computer with the assistance of reference material, and stereo-metric techniques can be used to define strict spatial survey areas (Harvey et al. 2010).

Like UVC, stereo-DOV has been widely used to detect spatial and temporal differences in the structure of fish communities (e.g., Watson et al. 2005). As the two methods have the ability to collect data over the same spatial area, information should be relatively comparable, allowing for a combination of methods within a single assessment program. The comparability of the data produced by UVC and single camera DOV has previously been demonstrated in studies conducted in large aquaria (Greene and Alevizon 1989; Bortone et al. 1991) and coral reefs on oceanic islands (Tessier et al. 2005; Pelletier et al. 2011). However, how UVC and stereo-DOV compare, and how this differs across a range of biogeographic regions (e.g., temperate rocky reefs and sub-tropical fringing reefs) where there are large variations in habitat complexity, water visibility, fish diversity, and abundance, is currently relatively unknown. Certainly, Langlois et al (2010) found changes in the comparability of other fish assemblage assessment methods when transitioning between tropical and temperate locations, which suggests that latitudinal position may be an important factor when considering methodologies. Similarly, as resource availability and the expense of obtaining information is likely to differ greatly between study locations and the agencies tasked with collecting data, it is essential to understand how comparable the financial and time costs of the methods are before decisions are made on which method is most suitable. The use of stereo-DOV also enables the incorporation of length assessment comparisons between methods, particularly for target species that have direct relevance to fisheries and recreational use management.

This study aims to assess the comparability of fish assemblage datasets produced by both UVC and stereo-DOV methodologies across a latitudinal gradient of shallow marine ecosystems along the coast of Western Australia. Surveys were conducted in tropical and sub-tropical high diversity systems and temperate mid-diversity systems to examine how changes in fish assemblage and environmental characteristics influence the respective results. The relative time (seen as a realistic metric of cost) required to complete surveys using each method was also assessed using detailed time budgets throughout the data collection and analysis process at each of the survey locations.

Materials and procedures

Study locations

Fish assemblages were surveyed at four widely separated locations along the western coast of Australia, between December 2010 and July 2011: Rowley Shoals Marine Park, Ningaloo Marine Park, Jurien Bay Marine Park, and Rottnest Island Marine Reserve (Fig. 1). The four locations cover approximately 15° of latitude, incorporating tropical and sub-tropical coral reefs, and temperate rocky reef systems that characterize the shallow reef habitats of this coast.

The Rowley Shoals Marine Park (S 17° 18.381′ E 119° 20.138′) lies within a high diversity (~528 fish species; Allen and Russell 1986) off-shore tropical coral atoll system. The shoals are characterized by mid/high coral cover and high visibility (ca 10-20 m). Nine sites were surveyed at this location, all situated on the northeastern outer slope of the shoals in 8-12 m.

The Ningaloo Marine Park (S 22° 26.066′ E 113° 38.827′) encompasses a sub-tropical fringing coral reef system on the mid-western coast of Australia. The system is characterized by variable visibility (ca. 5-15 m) and high fish diversity (~500

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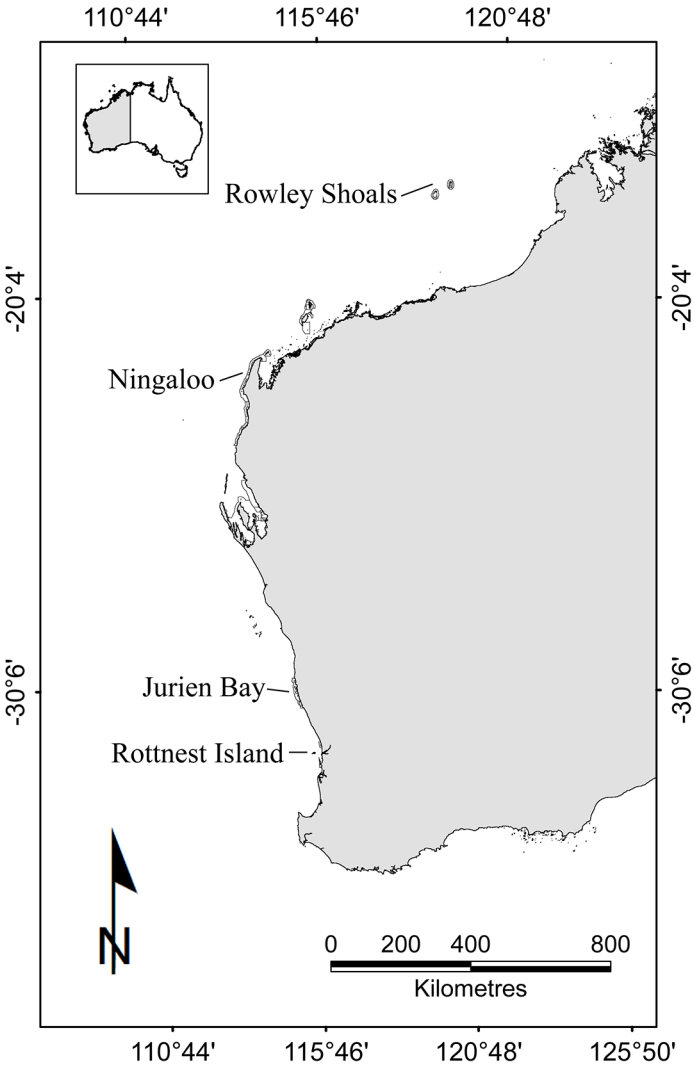

Fig. 1. Location of study sites (Rowley Shoals, Ningaloo, Jurien Bay and Rottnest Island) along the coast of Western Australia.

species; Hutchins 1994). Coral cover in the back-reef is variable, consisting mainly of plating, branching, and corymbose acroporids. Nine sites were surveyed at this location, all situated on the shallow back-reef in 2-6 m.

The Jurien Bay Marine Park (S 30° 25.741′ E 115° 1.214′) is a temperate marine system situated on the southwestern coast of Australia. Habitat within the park includes large areas of mid-complexity rocky limestone reef among a macro-algae dominated benthos. Visibility is generally mid-low (ca. 3-12 m), and fish diversity is lower than at the northern locations (~ 250 species; Hutchins 1994; Burt and Anderson 1997). Seven sites were surveyed at this location, all on shallow mid-complexity rocky reef substrate in 4-10 m.

The Rottnest Island Marine Reserve (S 32° 0.399′ E 115° 31.127′) is a temperate marine system situated approximately 17 km off the south-western coast of Australia. Habitat within the park is mostly mid-complexity rocky-limestone reef, with the predominant benthos being macro-algae. Visibility is variable (ca. 5-15 m). Rottnest Island receives a significant tropical water influence due to a strong offshore tropical Leeuwin Current (Cresswell and Golding 1980), with ~98 of the ~360 fish species present in the park classified as tropical (Hutchins and Pearce 1994). Six sites were surveyed at this location, all being on shallow rocky reef substrate in 5-10 m.

Survey procedure

Surveys of fishes were conducted at each location on SCUBA using two methods simultaneously: Underwater Visual Census (UVC) and stereo–Diver Operated Video (stereo-DOV). UVC was conducted by well-trained observers, highly experienced with the fish community present at each location. Surveys were conducted with a minimum water visibility of 5 m to ensure that the full transect width was easily visible to the UVC observer and on the stereo-DOV video. The stereo-DOV units consisted of two high definition Sony HG21 digital video cameras mounted in stereo configuration on a neutrally buoyant metal bar. The cameras were held in underwater housings and angled inwards at approximately 8°, resulting in the maximum field of view (Harvey and Shortis 1996). A synchronizing diode was positioned approximately 0.8 m in front of the rest of the unit, within both cameras' field of view, to facilitate synchronization of camera frames during the video analysis process. Field operation of the stereo-DOV was conducted by trained operators using a standardized approach adopted by the statewide marine monitoring group (Western Australian Marine Monitoring Program). The stereo-DOV operator swam along the transect line at a height of approximately 0.7 m above the substrate. The stereo-DOV cameras faced forward along the transect line at all times, avoiding swinging the cameras to include fishes that may not have been detected in the forward facing field-of-view. This standardized technique has been adopted to decrease bias associated with different operators using slightly different search patterns with the stereo-DOV units.

Before surveys, decisions were made regarding which species groups to include within the community suite at each location. Given the lower habitat complexity, diversity and abundances at Jurien Bay and Rottnest Island in the south, it was reasonable that all species could feasibly be recorded (with the exception of cryptic epifaunal species). Cryptic and nocturnal species (belonging to the families Apogonidae, Blennidae, Cirrhitidae, Gobiidae, Holocentridae, Pempheridae, Pinguipedidae, Plotosidae, Pseudochromidae) were excluded from surveys at Ningaloo and the Rowley Shoals as the higher habitat complexity, community diversity, and overall abundances prevented representative sampling of these groups using either the described UVC or stereo-DOV methodologies. Juveniles of all species were excluded from surveys at all locations.

Five replicate 50 m belt transects were conducted at each site in the Rowley Shoals, Ningaloo and Jurien Bay. Due to fragmented reef structure and the smaller size of prospective survey sites, five replicate 25 m belt transects were conducted

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at Rottnest Island. At all sites, a single UVC observer and stereo-DOV operator swam along each transect side by side at a constant speed. The UVC observer recorded the identity and abundances of all fish from the predetermined group list. Where identification to species level was not feasible, individuals were identified to the best possible level (e.g., Genus, Family) and recorded as 'unknown' at levels of higher taxonomic resolution. In addition, at the Ningaloo and Rowley Shoals locations, the UVC observer recorded the Total Length (TL mm) of selected serranid, lethrinid, lutjanid, and labrid species that are considered targets of fishers at that location (Sumner et al. 2000, pers. comm., WA Department of Fisheries). Each transect took 5 (for 25 m transects) to 8 (for 50 m transects) min to complete in a single pass, with both divers remaining immediately beside each other at all times to ensure that the same area was surveyed. Although this speed is slightly slower than would normally be used for stereo-DOV surveys (approx. 5 min for a 50 m transect), this was considered a fair trade-off between methodologies to ensure that they remained directly comparable. At temperate locations (Jurien Bay and Rottnest Island), all fish species on the predetermined group list were recorded by the UVC observer on the initial pass, across a 5 m belt width. At tropical and sub-tropical locations (Ningaloo and Rowley Shoals), all mobile fish species from the predetermined group list were recorded by the UVC observer on the initial transect pass on a 5 m belt width. The UVC observer then made a return pass along each transect, recording all smaller, more site-associated species of the predetermined group list on a 1 m belt width. This difference in UVC methodologies reflected the differences in species diversity and relative abundances between locations.

Stereo-DOV imagery was downloaded in the field onto external hard disk drives and transferred back to the laboratory for analysis. Videos were analyzed by a trained video analyst in the laboratory following the completion of field activities. The analyst was experienced at video-based analyses in a variety of locations, but lacked field experience. The analyst was provided full access to reference material, as well as the option of referring unknown identifications to taxonomic experts. The UVC observers were however excluded from this process to avoid influencing identification decisions based on their previous experiences on the same transects. All individuals observed within the 5 m transect belt on the videos were identified to the highest possible taxonomic level. Where identification to species level was not feasible, individuals were identified to the best possible level (e.g., Genus, Family) and recorded as 'unknown' at all levels of higher taxonomic resolution. Videos were analyzed using the software EventMeasure Stereo v3.13 (SeaGIS). At Ningaloo and the Rowley Shoals, the TL of serranid, lethrinid, lutjanid, and labrid species that are targeted by extractive fishing activities in each location, were measured for TL using the stereo-video component.

Due to functional limitations of the version of EventMeasure software available at the time of the study, it was not possible to differentiate between transect belt widths when recording fish abundances from video footage. As such, those smaller, site-associated species recorded on 1 m transect belts using UVC at the Rowley Shoals and Ningaloo were initially recorded on a 5 m transect belt using the stereo-DOV method. This data were later converted back to a value equivalent to a 1 m belt using species-specific conversion factors calculated using a software function that records the horizontal position of the fish relative to the stereo-camera's during the fish length measurement process. It should be noted that more recent versions of the EventMeasure software removes this problem.

Time budget

Accurate time budgets were maintained for all activities associated with each methodology during surveys conducted at all four locations. Time budgets were measured in 'people minutes' (number of minutes spent by each person devoting time to the activity) and were restricted to time actually spent actively completing the task. As such, time not directly associated with the actual survey task (e.g., travel time, total time in field) was excluded. This avoids influencing the results based on logistical differences in accessing and working in the different locations and enables a direct comparison of the methods themselves. Time budgets were broken up into three categories: Pre-Field Time (e.g., equipment calibration, fish identification revision), In-Field Time (e.g., in-water data collection, equipment maintenance, video download), and Post-Field Time (e.g., data entry, video analysis).

Data analysis

To investigate differences in species richness and overall abundance of fish between the two methods, the paired UVC and stereo-DOV data sets were analyzed separately at each location using repeated measure permutational analysis of variance (PERMANOVA) in the PRIMER 6 statistical package (Anderson et al. 2008). Method (UVC or stereo-DOV) was entered as the repeated measure, as data were collected simultaneously on the same transect, and transect was nested within the site level. Data were transformed using a Modified Gower Log_{10} resemblance transformation (Anderson et al. 2006). Differences in the proportion of unknown fish identifications between methods were analyzed using analysis of variance (ANOVA). The proportion of unknown identification was assessed as the percentage of fish that could not be reliably identified at a specific taxonomic/functional level, and were analyzed separately at species, genus, family, and functional group levels. The site level was excluded from the model as there were multiple sites where there were no unknown identifications, precluding a calculation of variance. Data were Log_{10} transformed to meet the assumptions of normality and heterogeneity of variances.

Analyses comparing the fish assemblages recorded by each method were conducted at all locations using PERMANOVA (Anderson et al. 2008). Analyses were conducted at both species and functional group levels. Species were placed into

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functional groups based on body size and dietary information from Fishbase (Froese and Pauly 2012). Tests were run using a repeated measures approach, due to the paired nature of the data. A Modified Gower Log_{10} resemblance transformation was used for the functional group data and Modified Gower Log_2 resemblance transformation for the relative abundance of species (Anderson et al. 2006). Principal Coordinates analysis (PCO) was used to visualize multivariate data in two dimensions and a canonical analysis of principal coordinates (CAP) used to compare fish community assessments from the two methods. Fish species contributing to differences in methodology were identified based on the strength of their correlation with the CAP axis that separated the two methodologies. For the purpose of this study, species were considered to have contributed to differences between methods if their CAP axis correlation value was greater than 0.4. For both PERMANOVA and CAP analyses, all unknown identifications were removed from the datasets.

Assessment

Species richness and abundance

The patterns in species richness varied across the four study locations. Although there was no significant interaction detected between factors ($P > 0.05$), there was a significant difference in both method ($P < 0.006$) and site ($P < 0.001$) at the Rowley Shoals, Ningaloo, and Jurien Bay locations (Fig. 2a). At

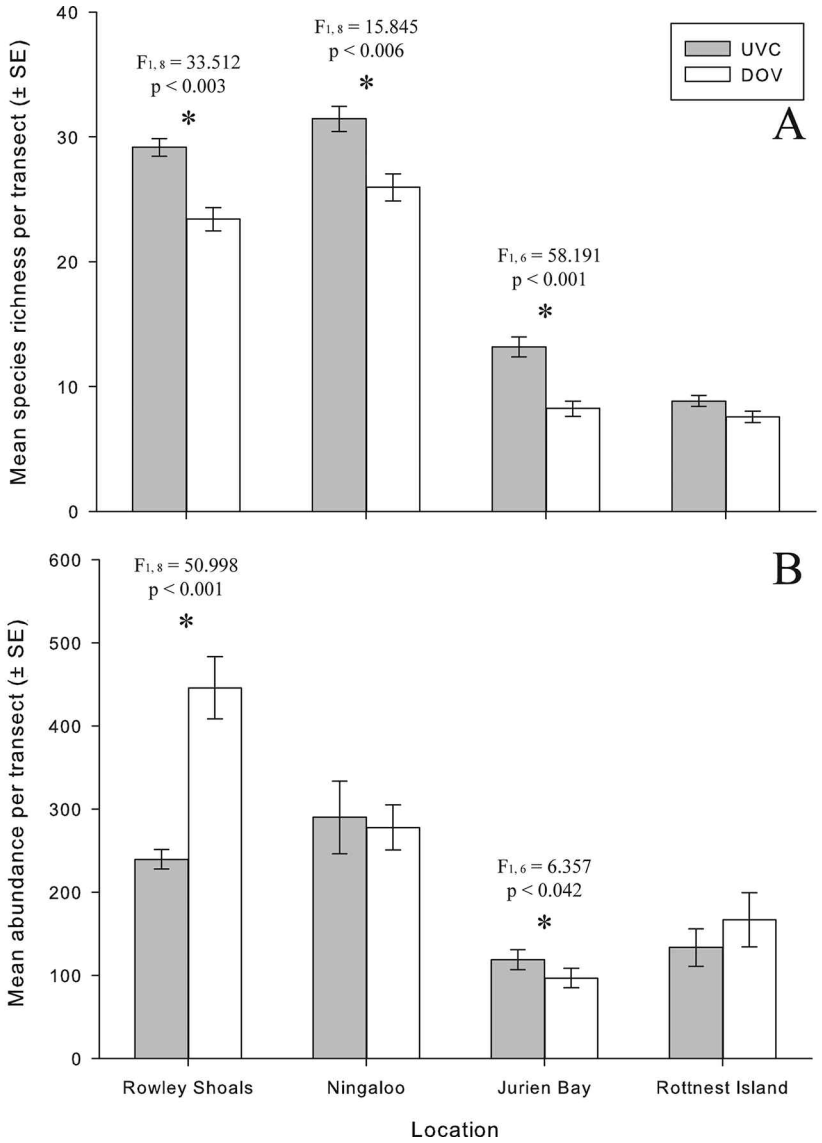

Fig. 2. Mean species richness (A) and abundance (B) per transect recorded using both UVC and stereo-DOV survey methods at the four study locations. Error bars are standard errors calculated from 30-45 transects at each location. Note that transects at the Rowley Shoals, Ningaloo, and Jurien Bay were 50 m in length, whereas transects at Rottnest Island were only 25 m in length. Statistics presented denote those locations where there was a significant difference between methods at $P < 0.05$.

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each of these locations, the UVC methodology recorded approximately five more species per transect than the stereo-DOV method. There was no difference in species richness estimates between methods or sites at Rottnest Island. Mean species richness was high overall at Ningaloo and the Rowley Shoals, with approximately 25-30 species observed on each transect. In comparison, Jurien Bay and Rottnest Island recorded only 10-15 species on each transect. It should be noted that data at Rottnest Island were collected using 25 m transects, while all other locations used 50 m transects, potentially underestimating the species richness values relative to the other locations.

There was a significant difference between the overall abundance of fishes between methods at the Rowley Shoals and Jurien Bay locations, and between sites at all four study locations (Fig. 2b). There was no interaction between method and site at any of the locations ($P > 0.05$). The stereo-DOV methodology recorded almost twice as many fish at the Rowley Shoals than the UVC method, although this difference was driven primarily by high abundance estimates of two species, *Pseudanthius tuka* and *Thalassoma amblycephalum*, on stereo-DOV transects. This pattern was reversed at Jurien Bay, with the UVC method recording marginally more fish overall than the stereo-DOV method. Mean abundances at the tropical Rowley Shoals and subtropical Ningaloo locations were 250-450 fish per transect, whereas the two temperate locations at Jurien Bay and Rottnest Island recorded mean abundances of < 200 fish per transect (Fig. 2b). Again, it should be noted that data for Rottnest Island were collected using 25 m transects, as opposed to the 50 m transects used at all other locations.

Assemblage composition

When assessed at the species level, there were significant differences in the recorded fish assemblages between methods at the Rowley Shoals, Ningaloo, and Jurien Bay locations (Table 1). Differences were most apparent at the Rowley Shoals where the PCO plots demonstrate that the samples representing the UVC and stereo-DOV species composition form two distinctly separate groups (Fig. 3). CAP analysis indicates these differences were mainly attributable to higher abundance estimates of mobile planktivores (e.g., *Thalassoma amblycephalum* and *Pseudanthis tuka*) and a single highly mobile scarid species (*Chlorurus sordidus*) when using stereo-DOV, and higher abundances of small bodied pomacentrids (e.g., *Chromis* spp. and *Pomacentrus vaiuli*) when UVC was employed (Fig. 3). Differences between the species assemblages recorded by each method are less obvious in the PCO plots for the Ningaloo and Jurien Bay locations. At Ningaloo stereo-DOV recorded higher abundances of the scarid, *Chlorurus sordidus*, but lower abundances of other scarid species, such as *Scarus chameleon*, *S. prasiognathus*, and *S. schlegeli* (Fig. 3). Estimates of small pomacentrids were also higher on UVC surveys at Ningaloo (e.g., *Plectroglyphidodon lacrymatus* and *Stegastes nigricans*) and Jurien Bay (e.g., *Parma occidentalis* and *Pomacentrus milleri*) and were a major driver of differences between methods at these locations (Fig. 3). Interactions between site and methodology at the Rowley shoals, Ningaloo, and Jurien Bay suggest that the

Table 1. Results from PERMANOVA tests assessing community assemblage at species and functional levels.

	Species			Functional groups		
	df	Pseudo F	p	df	Pseudo F	p
Rowley Shoals						
Method	1	9.0	0.002	1	10.9	0.002
Site	8	5.3	0.001	8	4.2	0.001
Method × Site	8	1.5	0.001	8	1.3	0.076
Error	72			72		
Ningaloo						
Method	1	5.4	0.001	1	10.2	0.004
Site	8	6.5	0.001	8	9.5	0.001
Method × Site	8	1.5	0.001	8	1.0	0.413
Error	72			72		
Jurien Bay						
Method	1	2.9	0.022	1	12.1	0.004
Site	6	7.8	0.001	6	11.0	0.001
Method × Site	6	1.2	0.046	6	1.0	0.527
Error	56			56		
Rottnest Island						
Method	1	1.8	0.221	1	1.6	0.243
Site	5	5.4	0.002	5	5.1	0.001
Method × Site	5	0.5	0.995	5	0.4	0.995
Error	48			48		

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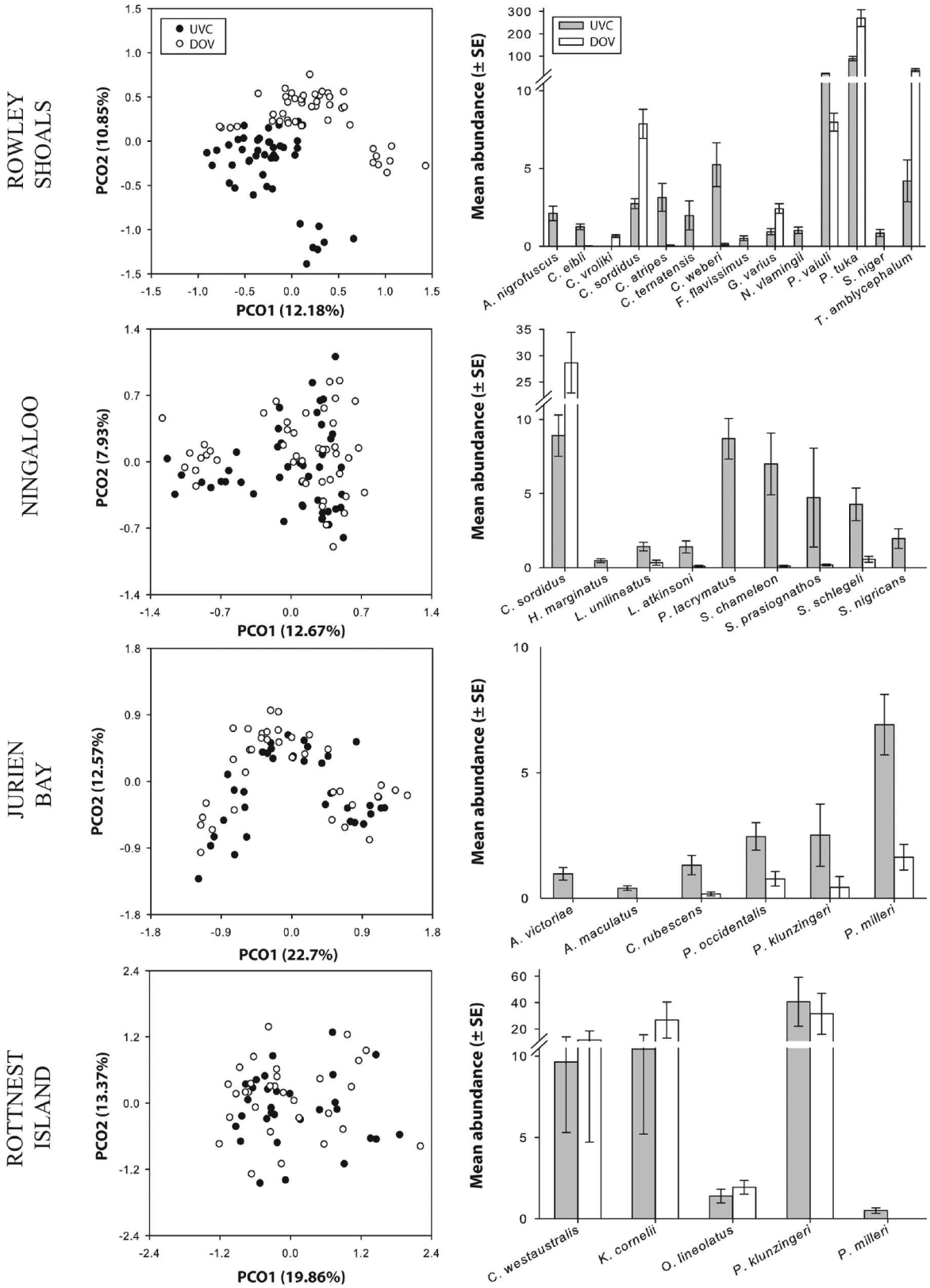

Fig. 3. Fish species assemblages assessed using UVC and stereo-DOV at the Rowley Shoals, Ningaloo, Jurien Bay, and Rottnest Island. Species level data presented as principle co-ordinate bi-plot, and histogram displaying the mean abundances per transect of those species responsible for a majority of variation between the UVC and stereo-DOV data sets at each location (defined as those species with a CAP axis correlation > 0.4).

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extent of differences attributable to methodology varied spatially (Table 1). The interactions were primarily driven by spatial differences in species assemblage and the ability of the two survey methods to detect particular species. For example, at the Rowley Shoals, stereo-DOV surveys estimated higher abundances of *C. sordidus* and *T. amblycephalum* than UVC (Fig. 3). However, stereo-DOV abundance estimates of these species were only high on the most southerly shoal. Assessments of the fish species composition using stereo-DOV and UVC at Rottnest Island were similar, and there was no significant interaction detected between methodology and site (Table 1, Fig. 3).

There were also significant methodological differences in the composition of fish assemblages when data were pooled into functional groups (Table 1). While not always distinct, there is a separation of groups representing the UVC and stereo-DOV functional compositions in the PCO plots for the Rowley Shoals, Ningaloo, and Jurien Bay locations (Fig. 4). At the Rowley shoals these differences were attributable to greater numbers of planktivores and detritivores when using stereo-DOV, whereas at Ningaloo the abundance of small croppers was greater using UVC than for stereo-DOV (Fig. 4). At Jurien Bay, estimates of small bodied croppers, invertivores, omnivores, and planktivores were all greater when using UVC. Methodological differences in assessing relative abundance of functional groups did not vary among sites and estimates of functional group abundance using stereo-DOV and UVC at Rottnest Island were very similar (Table 1).

The mean proportion of unknown identifications varied with respect to method and location at all four taxonomic/functional levels (significant interaction between method and location, $P < .007$). When examined in closer detail, differences between the UVC and stereo-DOV methodologies were most apparent at the species, genus and functional group taxonomic/functional levels, and at the Rowley Shoals, Ningaloo, and Jurien Bay locations ($P < 0.001$; Table 2). In all of these cases, the stereo-DOV recorded significantly higher proportions of unknown identifications than UVC, with the highest mean values occurring at the species level at the Rowley Shoals (15.19 ± 2.02) and subtropical Ningaloo (17.5 ± 1.94) locations. Proportions were lower at genera and functional group levels, and very few fish (<1%) could not be identified to family level using either method. It should be noted that the proportion of unknown identifications at the Rowley Shoals using UVC is solely attributed to an organizational protocol that groups *Ctenochaetus striatus* and *C. strigosus* into a single group called C*tenochaetus* spp.

Target species size estimates

The UVC method recorded higher abundances of targeted fish species than the stereo-DOV method at the Rowley Shoals and Ningaloo and length estimates were obtained for a higher percentage of these fish at both locations (Table 3). At the Rowley Shoals, the UVC methodology recorded 118 target individuals, obtaining length estimates for ~85% of these fish. In contrast, the stereo-DOV method recorded only half the number of target individuals as UVC at this location and length estimates were obtained for only half of these fish. Differences in these abundance estimates were mainly due to stereo-DOV's recording fewer lutjanids (e.g., *Aphareus furca* and *Lutjanus bohar*), whereas differences in the proportion of length measurements can be attributed to stereo-DOV's effectively measuring the length of fewer serranids (primarily *Cephalopholis argus*). Similarly, at Ningaloo, less than half the target fish recorded using the stereo-DOV method could be accurately measured (Table 3). In comparison, the UVC method recorded ~4 times as many target individuals, with length estimates being obtained for all of these fish. Major differences between the two methods at Ningaloo were due to UVC recording higher abundances of lethrinids (primarily *Lethrinus atkinsoni*, *L. nebulosus* and *L. punctulatus*) and providing proportionally more length measurements for lethrinids and serranids (primarily *L. atkinsoni*, *L. nebulosus*, *Cephalopholis argus*, *Epinephalus* spp., and *Plectropomus* spp.) than the stereo-DOV method.

Time budgets

The stereo-DOV methodology took more time to complete than the UVC methodology at all four study locations (Table 4). Rottnest Island had the greatest time ratio difference between methods with the stereo-DOV method taking ~3 times longer to collect and process data than the UVC method. It should be noted that surveys at this location were done with fewer sites (six) and using a smaller transect size (25 m belt) than all other locations. Despite the smaller transect lengths, stereo-DOV analysis times per site were similar to those for Jurien Bay where transects were 50 m in length (mean 204 min and 244 min per site for Rottnest Island and Jurien Bay, respectively). This can be explained by the comparably higher overall abundance of fish at Rottnest Island, which likely equated to an increase in video analysis time. At other locations, stereo-DOV took ~2-2.5 times longer than UVC. Post-Field time accounted for the majority of the total time differences between the two methods at all locations, video conversion, and processing (stereo-DOV) taking 3-7 hours longer per site than data entry from UVC. This difference was most pronounced at the Rowley Shoals and Ningaloo. Conversely, the UVC method required more time to collect data in the water, particularly at the tropical and subtropical locations (Rowley Shoals and Ningaloo) where each site took ~1 hour of extra time when using UVC. This is because each UVC transect was surveyed twice at the Rowley shoals and Ningaloo, once to estimate abundance of large mobile species and a second time to estimate abundance of smaller site attached fish (*see* "Materials and procedures").

Discussion

The comparability of stereo-DOV and UVC datasets was demonstrated to be dependent on both the complexity of the community in which assessments were made (i.e., species

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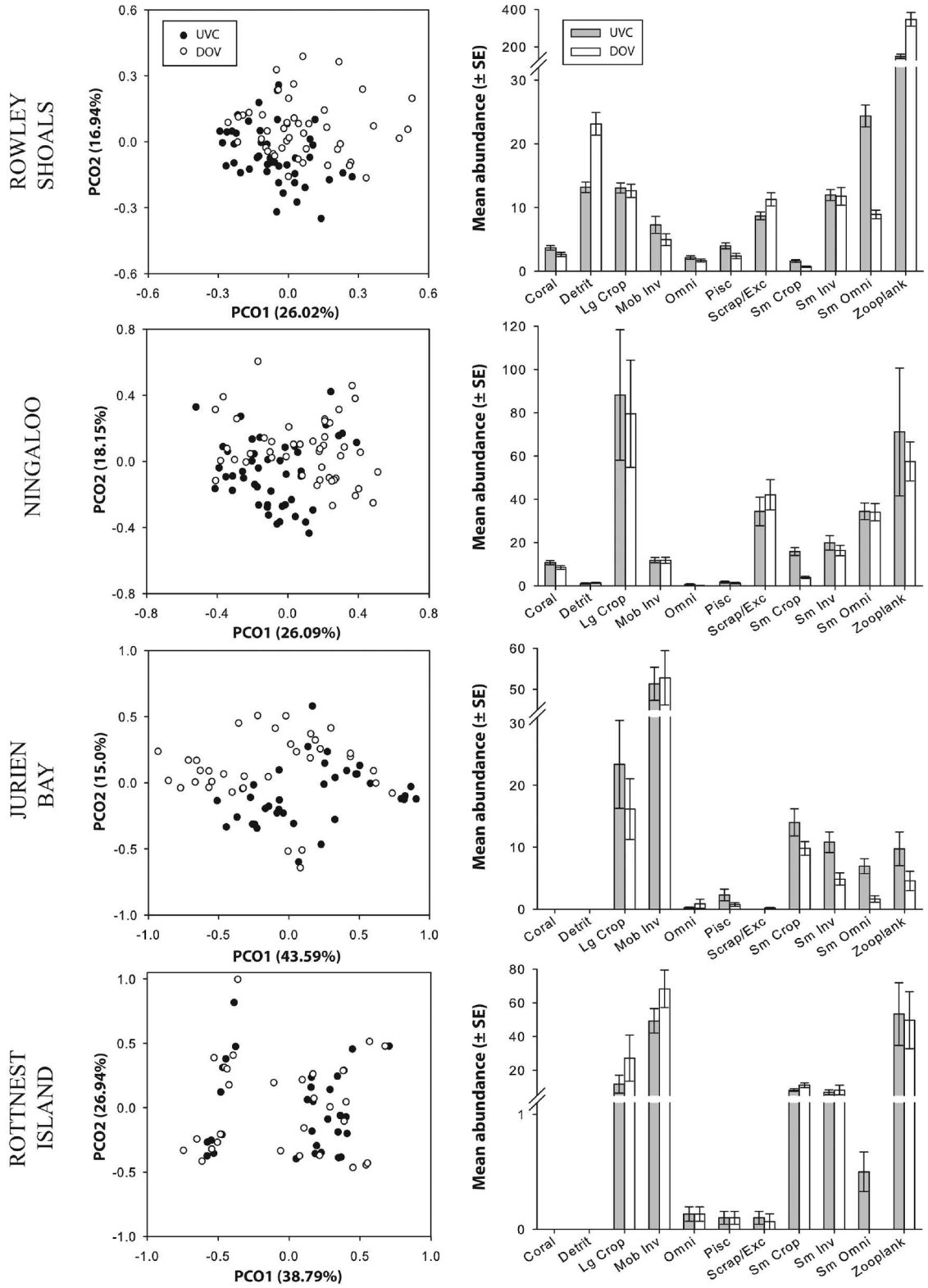

Fig. 4. Fish functional group assemblages assessed using UVC and stereo-DOV at the Rowley Shoals, Ningaloo, Jurien Bay, and Rottnest Island. Functional group level data presented as principle co-ordinate bi-plot, and histogram displaying the mean abundances per transect of all functional groups at each location. Functional group abbreviations on histogram axis are: Coral (Coralivore), Detrit (Detritovore), Lg Crop (Large Cropper), Mob Inv (Mobile Invertivore), Omni (Omnivore), Pisc (Piscivore), Scrap/Exc (Scraper/Excavator), Sm Crop (Small Cropper), Sm Inv (Small Invertivore), Sm Omni (Small Omnivore), and Zooplank (Zooplanktivore).

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Table 2. Mean proportions of unknown fish identifications made using the UVC and DOV methodologies at differing taxonomic/functional levels for each of the four study locations. Presented statistics derive from method level of ANOVA's used to analyze datasets at each taxonomic/functional level and the subsequent Tukey's HSD post-hoc tests.

Taxonomic/Functional level	Location	UVC Mean (± SE)	stereo-DOV Mean (± SE)	Tukey's HSD *P* value
Species ($F_{1, 302}$ = 292.92, $P < 0.001$)	Rowley Shoals	6.44 (± 0.58)	15.19 (± 2.02)	<0.001
	Ningaloo	1.74 (± 0.44)	17.5 (± 1.94)	<0.001
	Jurien Bay	0	9.61 (± 1.46)	<0.001
	Rottnest Island	0	2.12 (± 0.9)	0.105
Genus ($F_{1, 302}$ = 501.108, $P < 0.001$)	Rowley Shoals	0	6.48 (± 1.43)	<0.001
	Ningaloo	0	11.22 (± 1.4)	<0.001
	Jurien Bay	0	7.38 (± 1.07)	<0.001
	Rottnest Island	0	2.12 (± 0.9)	0.047
Family ($F_{1, 302}$ = 6.843, $P = 0.009$)	Rowley Shoals	0	0.01 (± 0.01)	0.999
	Ningaloo	0	0.023 (± 0.02)	0.999
	Jurien Bay	0	0.32 (± 0.16)	<0.001
	Rottnest Island	0	0	1.000
Functional Group ($F_{1, 302}$ = 398.192, $P < 0.001$)	Rowley Shoals	0	6.18 (± 1.43)	< 0.001
	Ningaloo	0.16 (± 0.05)	9.47 (± 1.4)	< 0.001
	Jurien Bay	0	7.48 (± 1.08)	< 0.001
	Rottnest Island	0	2.12 (± 0.9)	0.069

Table 3. Recorded abundance and number measured of target fish groups at the Rowley Shoals and Ningaloo using both stereo-DOV and UVC methods.

	Stereo-DOV			UVC		
	Recorded abundance	Measured	Proportion measured	Recorded abundance	Measured	Proportion measured
Rowley Shoals						
Labridae spp.	0	0	N/A	4	4	100
Lethrinidae spp.	0	0	N/A	0	0	N/A
Lutjanidae spp.	13	8	61.5	49	33	67.4
Seranidae spp.	51	22	43.1	65	64	98.5
Total	64	30	46.9	118	101	85.6
Ningaloo						
Labridae spp.	16	9	56.3	15	15	100
Lethrinidae spp.	25	6	24	122	122	100
Lutjanidae spp.	8	7	87.5	8	8	100
Serranidae spp.	5	1	20	16	16	100
Total	54	23	42.6	190	190	100

richness, overall abundances), and the taxonomic/functional level at which data were examined. Datasets were more comparable in low diversity temperate locations and when data were assessed at the broad functional level. This finding is consistent with previous studies that show diversity estimates of fish assemblages made by divers on coral reefs are higher than those made from assessments of video (Tessier et al. 2005; Pelletier et al. 2011), but in aquaria where diversity is low and community composition is known the two methods provide similar results (Greene and Alevizon 1989; Bortone et al. 1991).

When data collected from the two methodologies are scrutinized at the species level, differences between the techniques are primarily driven by highly abundant smaller bodied species (e.g., *Thalassoma amblycephalum* and *Pseudanthis tuka*), or those whose behavior or coloration make accurate identification difficult from video footage. Such differences are emphasized by the greater numbers of small, drably colored

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Table 4. Time budgets (total minutes devoted to each activity at each location) for stereo-DOV and UVC methods during surveys conducted at four study locations. Note that transects at the Rowley Shoals, Ningaloo, and Jurien Bay were 50 m in length, whereas those at Rottnest Island were 25 m in length. *n* = number of transects.

	Rowley Shoals (*n* = 45)		Ningaloo (*n* = 45)		Jurien Bay (*n* = 35)		Rottnest Island (*n* = 30)	
	DOV	UVC	DOV	UVC	DOV	UVC	DOV	UVC
Pre-Field	235	120	195	240	140	240	210	120
Equipment calibration	160	0	150	0	110	0	120	0
Equipment check and setup	75	0	45	0	30	0	90	0
Equip prep and ID update	0	120	0	240	0	240	0	120
In-Field	834	1191	981	1315	774	746	428	390
In-water data collection: total (per five transects)	630 (70)	1162 (129)	790 (88)	1260 (140)	608 (88)	706 (101)	288 (48)	360 (60)
Daily equip. maintenance	54	24	51	45	48	30	45	25
Video download	135	0	95	0	98	0	75	0
Equip. check and pack-up	15	5	45	10	20	10	20	5
Post-Field	5474	1530	4697	1100	1709	245	1224	160
Data entry and ID checks: total (per five transects)	0	1530 (170)	0	1100 (122)	0	245 (35)	0	160 (27)
Video processing: total (per five transects)	5474 (608)	0	4697 (521)	0	1709 (244)	0	1224 (204)	0
Total	6543	2841	5873	2655	2623	1231	1862	670

site-attached species recorded for UVC such as *Chromis atripes*, *C. weberi*, and *Plectroglyphidodon lacrymatus* at tropical and subtropical locations and *Parma occidentalis* and *Pomacentrus milleri* at temperate locations. In another example, the stereo-DOV method recorded much higher abundances of the mobile scarid *Clorurus sordidus* at the Rowley Shoals and Ningaloo locations, but much lower abundances of other scarid species such as *Scarus chameleon*, *S. prasiognathus*, and *S. schlegeli*. It is likely that this discrepancy is a result of differing identification between methods due to issues associated with differentiating between the similarly coloured initial phases of these species. Given that total abundances using the two methods are similar, lower species richness estimates when using stereo-DOV most likely relate to the high proportion of fish that cannot be positively identified to species level when viewed on video. Limitations associated with video image resolution, frame rate, and over/under-exposure, combined with limited time that fish are often within the camera's field of view means that fish presence is often noted, but identity cannot accurately be resolved to the species level (Cappo et al. 2003). Conversely, divers carrying out UVC are able to focus upon and follow fish of questionable identity to assist in species verification (Bortone et al. 1991).

Although overall abundances were similar at two of the four locations, the UVC method recorded a higher abundance of larger bodied species commonly targeted by fishers at the tropical locations than the stereo-DOV method (e.g., lethrinids, lutjanids, serranids). This has major implications, as the questions frequently asked of fish datasets commonly revolve around spatial or temporal differences in the abundance or biomass of these groups. If not effectively accounted for during cross-method comparisons, this bias has the potential to alter or mask any real change in these metrics. Lower estimates of target species from the stereo-DOV technique partially relates to the relatively static field of view of the video camera and behavior of many target fish species. For example, one of the species responsible for differences in stereo-DOV and UVC estimates at the Rowley Shoals, *Aphareus furca,* is often observed mid-water (Myers 1999), and are less likely to be detected by a camera with a field of view directed just above the benthos. Moreover, some fish species frequently swim ahead of the divers, toward the edge of the diver's visibility range. The human eye can detect these fish, but the more restrictive video imagery (associated with correct exposures, contrast, etc.) commonly does not perceive these fish, particularly if the visibility is low. Similarly, lower abundance estimates of serranids on stereo-DOV surveys is likely due to the limited ability of the camera to effectively search the highly complex coral reef habitat where many of these semi-cryptic species reside. Divers are, however, more likely to include fish in surveys that were observed outside the transect (Harvey et al. 2004), which may have contributed to higher estimates of target species when using UVC. This is less of a problem with stereo-DOV as transect dimensions can be accurately assessed to ensure only fish within transect boundaries are included.

One apparent advantage of the stereo-DOV system is that it measures fish size more accurately and precisely than divers carrying out UVC, which provides greater power to detect change in fish length and biomass (Harvey et al. 2001a, 2001b). However, fish bodies are often partially obscured by other objects in videos (e.g., other fish, benthos), or the entire

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length is only visible in one of the stereo-cameras, preventing accurate measures of body dimensions using the computer software. Furthermore, stereo-DOV measurements of fish length are most effective when fish are observed at an angle < 75° from the stereo cameras (Harvey et al. 2010) and fish that are recorded at higher angles relative to cameras are often not measured (e.g., Harvey et al. 2002). Consequently, UVC tended to obtain measurements for 2-4 times more target fish than the stereo-DOV method. Despite the lower detection, this is sufficient replication to obtain the same power to detect changes as when using stereo-DOV to obtain fewer but more precise length estimates (Harvey et al. 2001a, 2001b). Importantly, while UVC may be a more powerful tool in detecting spatial and temporal changes in abundance, these results infer that both methods may have similar capacity to detect spatial and temporal changes in biomass for species most at risk from extractive activities.

The UVC method has been popular among researchers and monitoring programs because it provides nondestructive estimates of fish abundance that require little investment in equipment and hardware (e.g., cameras, high-spec computers) and computer software (Murphy and Jenkins 2010). The UVC method was also found to capture and collate data 2-3 times faster than stereo-DOV. This is primarily due to the additional time required to process and analyze video footage from stereo-DOVs after the in-field work component has been completed. The post-field time investment alone was found to be 3.5-7.5 times more costly for stereo-DOV than UVC in this study. Indeed a recent survey of scientists and managers indicated the time to process images was a major limitation to using video for monitoring and researching of fish (Harvey et al. 2011). The time associated with processing video may be reduced by only counting and measuring a sub-set of species or collating data at higher taxonomic and functional levels. If required, video can then be later re-analyzed to access more detailed or additional information (e.g., Wilson et al. 2009) or validate results (Langlois et al. 2010). This permanent record is a clear advantage of stereo-DOV over UVC. It is also possible that advances in computer software technology in the near future will facilitate the automation of much of the digital imagery analysis process (MacLeod et al. 2010). Although such technology is still being developed and tested, this may assist with bringing the time investment disparities between the methods closer together.

Time to process videos was particularly long at the Rowley shoals and Ningaloo where both species richness and abundance of fish was high. However, a greater richness and abundance of fish also requires more time to collect UVC data. UVC surveys at the Rowley shoals and Ningaloo took 70% to 80% longer to collect data than stereo-DOV and previous studies suggest stereo-DOV surveys diverse fish communities 2-5 times faster than UVC (Tessier et al. 2005; Pelletier et al. 2011). This may be important if field time is limited or an expensive component of the project.

Comments and recommendations

As with many methodologies stereo-DOV and UVC each have a set of unique attributes and limitations. Stereo-DOV provides a permanent record of fish communities, with the capacity to accurately and precisely measure the size of both fish and survey areas. Field data can be collected relatively easily by stereo-DOV operators with little experience in fish identification, and this can limit the potential for inter and intra observer variation during data collection (Harman et al. 2003; Watson et al. 2010). Video must be analyzed by personnel with some taxonomic knowledge, although captured images and repeated playback allow reference to field guides and expert taxonomists. Nonetheless, processing of videos is time-consuming, still requires a degree of observer expertise/training, many of the identifications are not verified by experienced taxonomists, and limitations imposed on the clarity of video imagery by current technological restrictions can result in fish not being observed, or only identified at coarse taxonomic levels (e.g., family). The heavy financial cost for the purchase/hire of the required hardware (e.g., digital video cameras, high quality computers, software, data storage) is also a factor that must be taken into account. Some of these limitations are however likely to ease in the future, as technological advancements continue to increase digital video imagery clarity and resolution to a quality more closely resembling that of the human eye, and the financial cost of basic hardware requirements declines.

Conversely, UVC identified more fish to a lower taxonomic level and provided abundance estimates of more species than the stereo-DOV method, including those most likely to be targeted by extractive activities. Additionally, UVC was able to obtain length measurements for a higher proportion of these target species. Data collection using UVC does not require the use of expensive equipment and the time required to enter information into a database is much less than that taken to analyze videos. UVC does however require divers to have extensive taxonomic knowledge, and once data are collected, it cannot be validated. Clearly the choice of an appropriate method to survey fish should be based on project objectives and availability of resources. Where appropriate expertise is available, UVC would be a more cost-effective choice. However, where expertise is lacking or is likely to be inconsistent, and sufficient resources and funding are available, stereo-DOV may be a more appropriate option. As all methods have some intrinsic limitations, a combination of techniques may be the best strategy in some situations to obtain appropriate community measures (Connell et al. 1998).

Our study demonstrates that there is a degree of variability in the comparability of datasets obtained using stereo-DOV and UVC methods. Whereas datasets obtained in lower diversity temperate systems are relatively similar, particularly when examined at higher taxonomic/functional levels, those obtained in higher diversity tropical/subtropical systems are

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much more distinct. However, these differences can often be attributed to a small number of species/groups, suggesting that spatial and temporal comparisons of data collected using the two methods may be possible in some cases. To avoid potentially confounding effects of methodology, it is suggested that correction factors be developed where possible, or that certain groups are excluded from any analyses that incorporate data from both methods.

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